



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AMARNATH FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

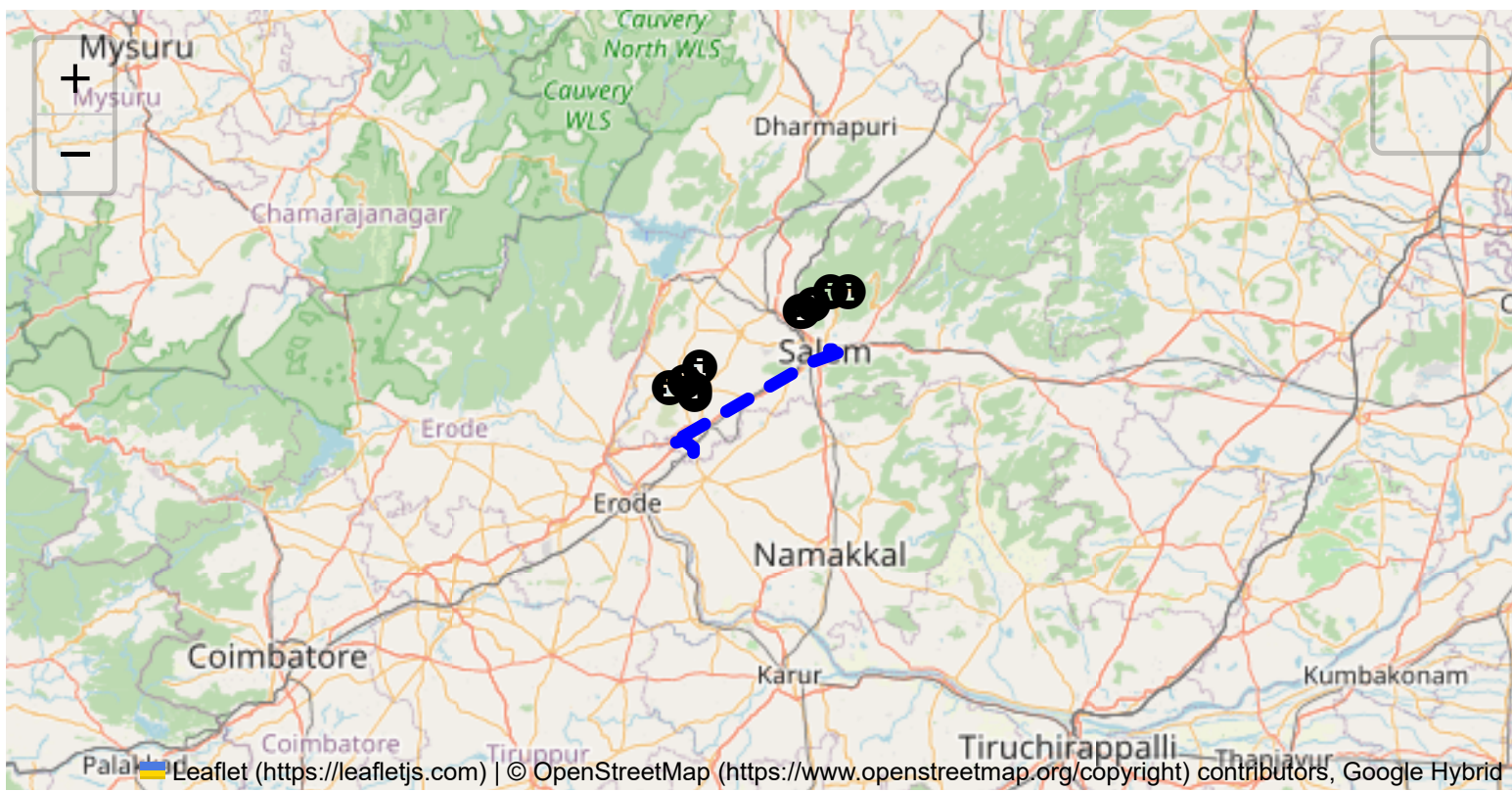
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

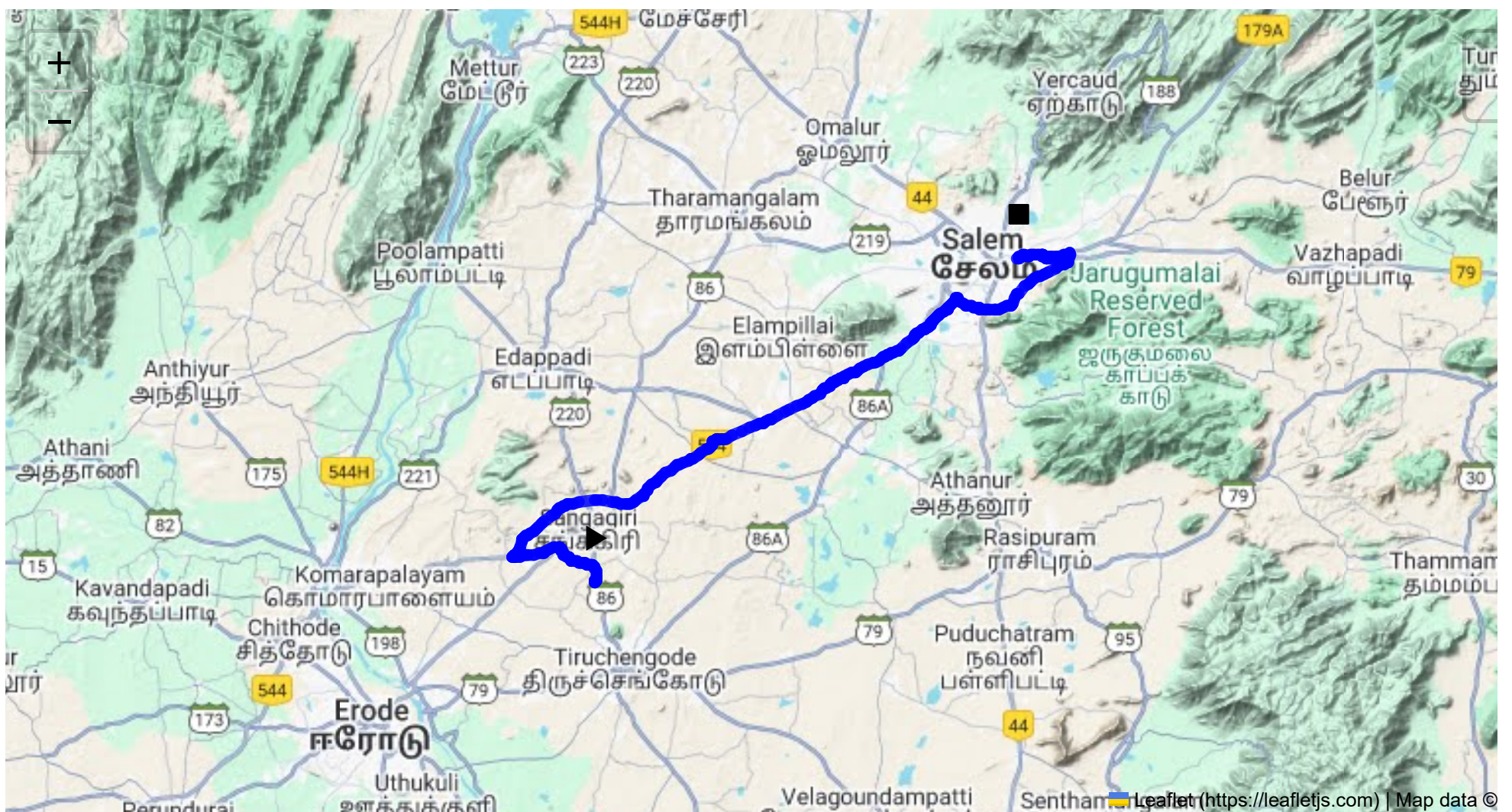
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 64.31 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.660097, 78.169144)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu to 105, Salai Rd, Ponnammampet, Salem, Tamil Nadu covers approximately 64.31 kilometers. The route primarily follows the NH544, passing through various locales such as Sankari RS, Sankari West, Padaiveedu, Neykkarapatti, Uthamasolapuram, and

Kondalampatti in Salem. The estimated travel time is around 1.33 hours for heavy vehicles carrying hazardous material.

Typical Weather Conditions

Tamil Nadu experiences a tropical climate. Weather along this route is typically hot and humid, with high temperatures most of the year. The monsoon season, from June to September, can bring heavy rainfall, leading to potential flooding and reduced visibility. Pre-monsoon showers may also occur, making roads slippery and hazardous.

Traffic Patterns

Traffic congestion is a common occurrence in the towns and cities along this route, particularly near Sangagiri and as you approach Salem. Peak traffic hours are typically from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM. These areas are prone to bottlenecks, especially on the NH544 near interchange points.

Road Quality and Infrastructure

The NH544 segment is in relatively good condition, being a national highway. However, certain stretches may have potholes and uneven surfaces, especially after the monsoon season. Smaller roads in towns like Sankari West and Padaiveedu may be narrower with mixed traffic, requiring caution.

Alternative Routes

In case of emergencies, one alternative is taking a detour via the SH86 at Sankari which reconnects to NH544 at a different segment before heading into Salem. This route may not be as direct but can bypass major congestion zones.

Local Regulations on Hazardous Material Transport

Tamil Nadu has specific regulations regarding the transport of hazardous materials, requiring appropriate permissions and adherence to safety protocols. Vehicles carrying such materials should have proper signaling, and drivers must be adequately trained.

Historical Incidents

There have been few recorded incidents involving heavy vehicles and hazardous materials on this route. However, intersections and sharp turns have been noted as critical points where accidents could occur, indicating the need for cautious driving.

Environmental Considerations

The regions along the route may have some ecologically sensitive areas, particularly where the road intersects with natural habitats. Noise pollution and emissions should be minimized, and accidental spillages must be avoided to prevent environmental damage.

Communication Coverage

The route generally has consistent mobile network coverage, but there are potentially weak zones as you travel through rural sections and under bridges. It's advisable to have alternative communication means such as a satellite phone for emergencies.

Emergency Response Times

Emergency response along the NH544 is relatively quick, estimated at 30-45 minutes due to proximity to major towns. However, in rural areas, response times may extend to over an hour, depending on the nature of the incident.

Risk Assessment Summary

Overall, the route from Sangagiri to Salem via the specified localities is moderately safe for transporting hazardous materials, provided that all protocols are followed. Major risks include weather-related hazards, road conditions in certain segments, and peak-hour congestion. Adequate preparations such as route planning, emergency communication setups, and adherence to local regulatory frameworks are essential to minimize risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.45847, 77.85140	30 KM/Hr
8	Turn	Medium	11.46048, 77.85036	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr
10	Turn	Medium	11.46314, 77.84928	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45634, 77.81673	15 KM/Hr
13	Turn	High	11.61233, 78.10397	15 KM/Hr
14	Turn	High	11.61216, 78.10423	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.63300, 78.12746	15 KM/Hr
16	Turn	Medium	11.65627, 78.20215	30 KM/Hr
17	Turn	Medium	11.65664, 78.20240	30 KM/Hr
18	Blind Spot	Blind Spot	11.66372, 78.20649	10 KM/Hr
19	Turn	High	11.66025, 78.18962	15 KM/Hr
20	Turn	Medium	11.66038, 78.18950	30 KM/Hr
21	Turn	High	11.66151, 78.18944	15 KM/Hr
22	Turn	Medium	11.66203, 78.18764	30 KM/Hr
23	Turn	Medium	11.66290, 78.18717	30 KM/Hr
24	Turn	Medium	11.66303, 78.18699	30 KM/Hr
25	Turn	Medium	11.66246, 78.17806	30 KM/Hr
26	Turn	Medium	11.66079, 78.16980	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
20	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
21	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
29	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium
33	hospital	S Palaniandi Mudaliar Memorial Hospital (Spmh Hospital)	11.661301, 78.1920085	30 km/h	Medium
34	hospital	Lakshmi Hospital	11.6601357, 78.1904563	30 km/h	Medium
36	pharmacy	MedPlus	11.6600134, 78.1888946	30 km/h	Medium
40	hospital	Dr. Malathi Anbalaya Hospital	11.6642243, 78.1754526	30 km/h	Medium
43	hospital	Sarvana Hospital Pvt. Ltd.	11.6624826, 78.1715561	30 km/h	Medium
45	hospital	Dr. Nalini Sridhar Hospital	11.6603637, 78.168978	30 km/h	Medium
47	hospital	Selvi Hospital	11.6588, 78.168423	30 km/h	Medium
48	hospital	Thamilchelvi Hospital	11.6587954, 78.1686235	30 km/h	Medium
49	hospital	K.N. Nataraj Hospital	11.6581476, 78.167265	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
50	hospital	Vinayaka Misson Hospital	11.658013, 78.1665941	30 km/h	Medium
51	hospital	Ac Hospitals	11.6578435, 78.1667691	30 km/h	Medium
52	hospital	KSB Speciality Hospital	11.6578993, 78.166628	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Holy Cross Matriculation Higher Secondary School	11.6635657, 78.203532	30 km/h	Medium
30	school	Govt. High School, K.N.Colony	11.6617684, 78.1970397	30 km/h	Medium
31	school	Jothi Matriculation School	11.6622074, 78.1968379	30 km/h	Medium
32	school	Salem Polytechnic. ITI	11.6599961, 78.1962722	30 km/h	Medium
35	school	Govt. Girls School, Ammapet	11.6605896, 78.1900977	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	school	Govt Boys School,Ammamet	11.6594455, 78.1874454	30 km/h	Medium
38	college	Sakthi Kailash Women's College	11.6640193, 78.1842566	30 km/h	Medium
39	college	AVS Engineering College	11.6634248, 78.1833888	30 km/h	Medium
41	college	IIHT	11.6627146, 78.1700314	30 km/h	Medium
42	school	Sri Vasavi Matric HR Sec School	11.6601017, 78.171237	30 km/h	Medium
44	college	IIHT	11.6632922, 78.1707003	30 km/h	Medium
46	college	Indian Instituite of Handloom Technology	11.6627734, 78.1698764	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

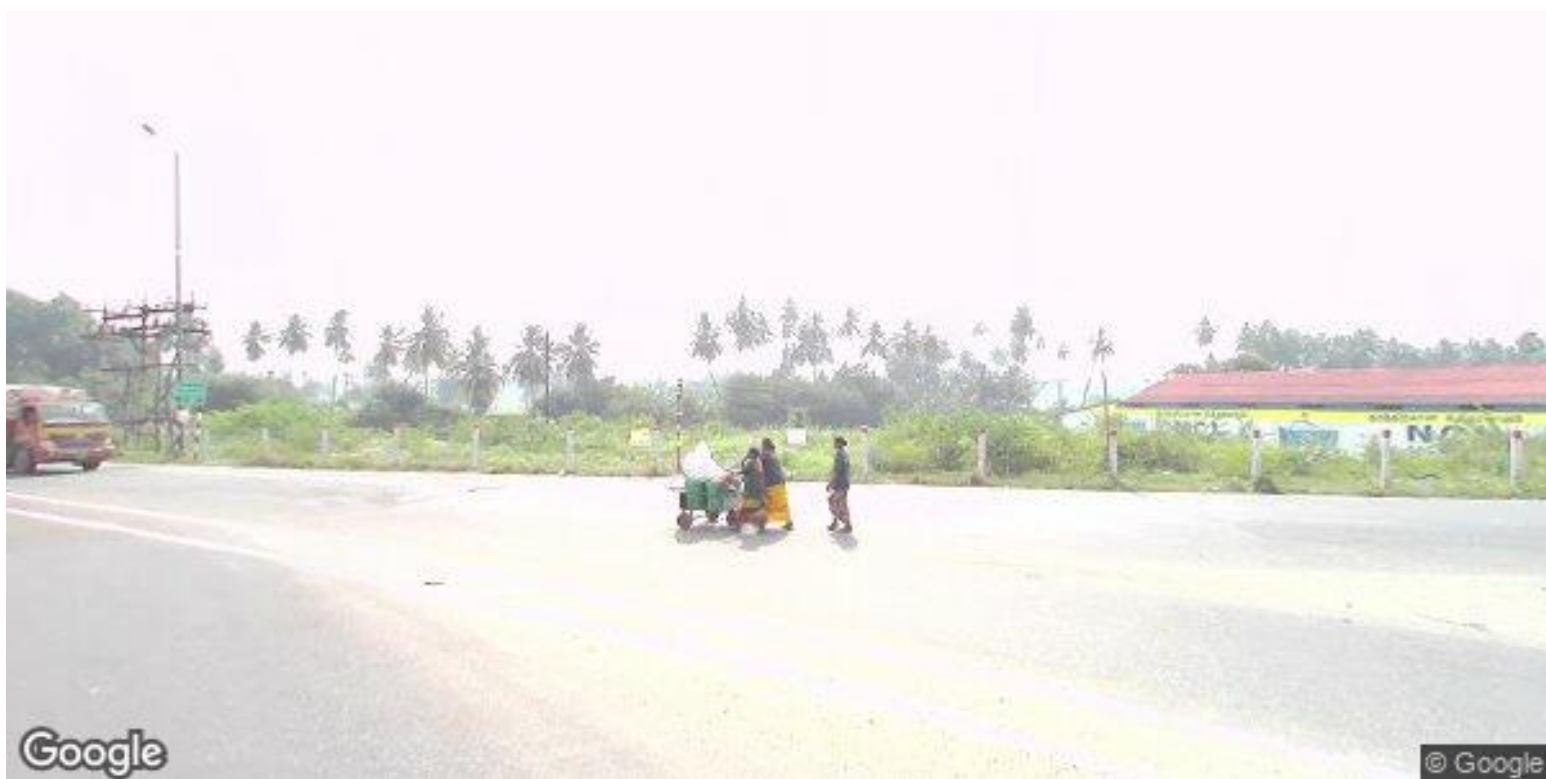


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65627, 78.20215



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65664, 78.20240



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.66372, 78.20649



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66025, 78.18962



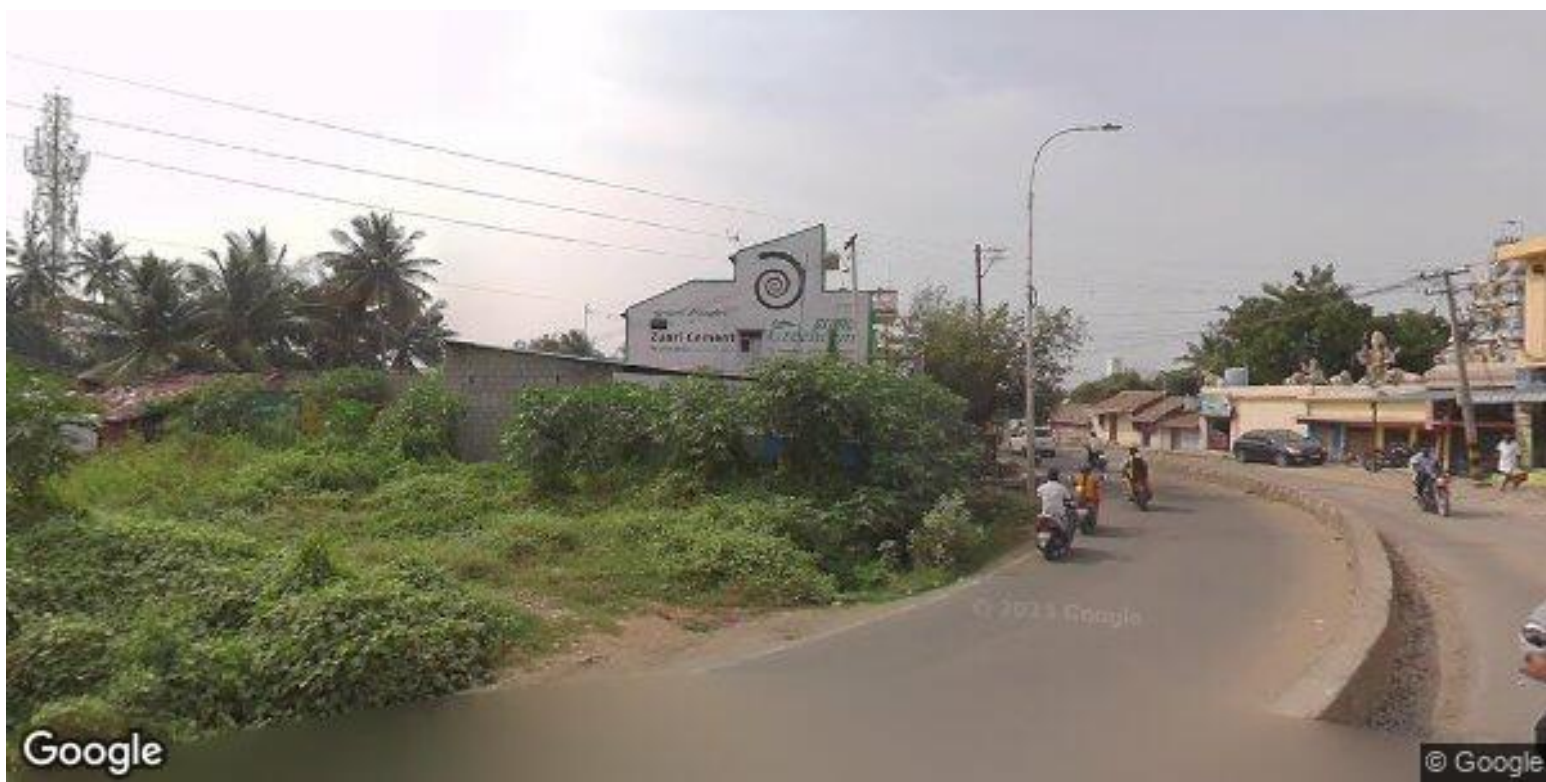
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66038, 78.18950



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66151, 78.18944



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66203, 78.18764



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66290, 78.18717



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66303, 78.18699



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66246, 78.17806



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66079, 78.16980
